	PARTICIPANT USE OF PROFICIENCY TESTING REPORT POLICY	
AzurESky "ensure quality of test"	Document no: ASMF 7.4.3	
	Compiled by:	**Sifiso Nyambi - Laboratory Manager**
	Approved by:	**Nonhlanhla Msibi - Operations Manager**

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Introduction

A laboratory implies all organizations performing testing or calibration activities, and this can extend to other inspection bodies and other PT scheme or RM producers. The basic use of PT for a laboratory is to assess its performance for the conduct of specific tests, measurements or calibrations. Participating in a PT scheme provides the laboratory with the opportunity to compare their results with other laboratories through an independent external assessment.

The results and information received from the participation in PT scheme will provide laboratories with either a confirmation that their performance is satisfactory, or an indication that there are potential problems and that corrections should be made. To maximise the benefits of participating in PT schemes, it is essential that participants pay attention to the documents provided by the Azuresky PT provider.

However, the use of PT should be much wider than the basic statement of whether the laboratory is competent or not. Laboratories can benefit from the participation in PT schemes in various ways.

1. Identifying measurement problems

If a laboratory's result in a PT scheme indicates unsatisfactory performance, this should start a process of investigation of potential sources of error. Without participation in the PT scheme, such sources of error could remain undetected and the laboratory would not have been able to undertake appropriate corrective actions. This, in turn, could have resulted in the laboratory continuing to provide poor results to its customers or other stakeholders. Eventually, such errors could also lead to the loss of reputation of the laboratory or to legal or other action being taken by the customers or other stakeholders, such as regulatory bodies. In this regard the use of PT may be considered to be a risk management and quality improvement tool.

2. Comparing measurement procedures

For some laboratories, their participation in a PT scheme might be used to trial their performance using a new, modified or infrequently conducted measurement. In other cases, the participation may provide an opportunity to compare the results achieved by the

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laboratory using different measurement procedures (or when determining different concentration levels, etc.) to those normally used by the laboratory.

The PT report might, in some cases, provide summaries and comparisons of all the measurement procedures or commercial kits used by the participants. For new or unusual activities, such data could be most valuable and assist the future selection of appropriate measurement procedures by the laboratory or indicate the need for additional investigation before adoption of new measurement procedures.

3. **Comparing operator capabilities**

If sufficient PT items are available to allow more than one operator within a laboratory to carry out the analysis, the laboratory has the added benefit of being able to compare the results of its operators. In addition, this might also provide some inputs to the laboratory's evaluation of its measurement uncertainty for the relevant measurements. This might also allow the laboratory to compare the between-operator precision with published (or otherwise available) data for the measurements concerned. Azuresky PT scheme might, in some cases, enable separate results of one or more operator to be reported. As more than one sample for general analysis PT items are distributed to participants.

4. **Comparing analytical systems**

PT results can provide an objective external assessment of the relative performance of analytical systems (on the same or different sites) used within a laboratory.

5. **Improving performance**

When a laboratory is not satisfied with its own results in a PT scheme, this provides an opportunity for the laboratory's management to investigate areas where its future testing could be improved. This might, for example, include additional operator training, adoption of new or modified measurement procedures, enhancing quality of data, or equipment modifications, calibration or replacement.

6. **Educating staff**

Azuresky PT scheme may be used to provide training to participants and empower the participant of information on measurement procedures, data interpretation, uncertainty assignments, etc., which arise from the overall results of the PT report.

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7. **Exchange of information with the PT provider**

Following the issue of a PT report, the laboratories usually have the possibility to contact the PT provider in order to obtain additional information about the results, or advice concerning the potential cause of non-satisfactory results. It may be necessary to hold "Participants' meetings", which can provide very useful information for the laboratories.

8. **Instilling confidence in staff, management, and external users of laboratory services**

Successful performance in a PT scheme can provide individual staff and their direct managers with additional confidence. Other management, including those without relevant technical expertise, can also be re-assured by the successful performance of their staff, often in areas of critical significance to their organization's activities and responsibilities. External users of laboratory services, including their customers and the parties affected by the outcome of measurement, can also be given added confidence when made aware that a laboratory is willing to have its performance regularly evaluated through PT participation. Therefore, Azuresky PT report is permissible to be shared by participant and satisfy the objective of providing confidence of their laboratory performance.

9. **Measurement uncertainty**

The laboratory's results from its participation in PT can, with caution, be used to check the evaluated measurement uncertainty, since that uncertainty should be compatible with the spread of results obtained by that laboratory over a number of PT rounds. The "PT approach" can, in specific cases, also be used to evaluate the uncertainty. For example, if the same measurement procedure is used by all the participants in the PT scheme, the standard deviation is equivalent to an estimate of the reproducibility and can, in principle, be used in the same way as the reproducibility standard deviation obtained from an ILC undertaken for the purpose of performance characterisation of the measurement procedure. Results from PT for sampling can be used to evaluate measurement uncertainty arising from between-sampler bias.

10. **Use of PT items as Internal Quality Control materials**

In some PT schemes, where there is sufficient, stable material provided to participants, the unused material could be used as a QC material for monitoring measurement performance as part of the laboratory's QC procedures. Where appropriate, the assigned values for the

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PT item might be considered useful as internal reference values for QC of measurement, operator training.

11. Determining measurement precision and/or trueness

Depending on the design, Azuresky PT scheme will be useful in determining the precision (repeatability and reproducibility) or comparative trueness of the measurement procedures used in the PT scheme. In most cases, determination of precision and trueness of the measurement procedures is not the primary aim of the PT scheme. To achieve this, further information is often needed and may be obtained from Azuresky.

12. Satisfying regulators and accreditation bodies

The successful performance of a laboratory in a PT scheme (or its effective correction of measurement problems after an unsuccessful performance) may provide regulators and accreditation bodies with confidence in the laboratories whose data they endorse or otherwise recognise. The clear benefit for the laboratories is the continuation of their standing as competent organizations. However, the internal benefits to laboratories, their staff and management, should be of most value if they view PT as a vital tool for ongoing maintenance of confidence and improvement, irrespective of whether the laboratory needs to participate for accreditation purposes.

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